

CLINICAL INVESTIGATION

Eye

LOCAL TUMOR CONTROL, VISUAL ACUITY, AND SURVIVAL AFTER HYPOFRACTIONATED STEREOTACTIC PHOTON RADIOTHERAPY OF CHOROIDAL MELANOMA IN 212 PATIENTS TREATED BETWEEN 1997 AND 2007

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Purpose: To evaluate long-term local tumor control, visual acuity, and survival after hypofractionated linear accelerator–based stereotactic photon radiotherapy in patients with choroidal melanoma.

Methods and Materials: Between 1997 and 2007, 212 patients with choroidal melanoma unsuitable for ruthenium-106 brachytherapy or local resection were treated stereotactically at a linear accelerator with 6-MV photon beams at the Medical University of Vienna in five fractions over 7 days. Twenty-four patients received a total dose of 70 Gy (five fractions of 14 Gy), 158 a total dose of 60 Gy (five fractions of 12 Gy) and 30 patients a total dose of 50 Gy (five fractions of 10 Gy) applied on the 80% isodose. Ophthalmologic examinations were performed at baseline and every 3 months in the first 2 years, every 6 months until 5 years, and once a year thereafter until 10 years after radiotherapy. Assessment of visual acuity, routine ophthalmologic examinations, and measurement of tumor base dimension and height using standardized A-scan and B-scan echography were done at each visit. Funduscopy and fluorescein angiography were done when necessary to document tumor response.

Results: Median tumor height and volume decreased from 4.8 mm and 270.7 mm³ at baseline to 2.6 mm and 86.6 mm³ at the last individual follow-up, respectively ($p < 0.001$, $p < 0.001$). Median visual acuity decreased from 0.55 at baseline to hand motion at the last individual follow-up ($p < 0.001$). Local tumor control was 95.9% after 5 years and 92.6% after 10 years. Thirty-two patients developed metastatic disease, and 22 of these patients died during the follow-up period.

Conclusion: Hypofractionated stereotactic photon radiotherapy with 70 to 50 Gy delivered in five fractions in 7 days is sufficient to achieve excellent local tumor control in patients with malignant melanoma of the choroid. Disease outcome and vision are comparable to those achieved with proton beam radiotherapy. Decreasing the total dose below 60 Gy seems to be possible. © 2011 Elsevier Inc.

Choroidal melanoma, Hypofractionated stereotactic photon radiotherapy, Local tumor control, Visual acuity, Melanoma-related death.

INTRODUCTION

Uveal melanoma is the most common primary ocular malignant tumor in adults, with an occurrence of 6 to 7 cases per million people per year (1, 2). Although in some patients enucleation is still required because of large size of the tumor at the time of presentation, eye-preserving approaches have shown to be equally successful regarding overall survival and metastasis-free survival (3, 4). Depending on tumor size and location, therapeutic strategies include radiotherapy, surgical techniques, or a combination of both (5–11).

Besides the development of successful surgical techniques, radiotherapy remains the standard eye-conserving

treatment of uveal melanoma. Episcleral plaques with ruthenium-106, iodine-125, or strontium-90 for smaller uveal melanoma are used for brachytherapy (12–17). Charged-particle therapy using protons (historically also with helium ions) has been established as an effective method for more than 30 years (18–24). This form of teletherapy is used with surgical application of the tantalum markers before radiotherapy and is currently available only in a limited number of specialized centers worldwide.

In the past decade, linear accelerator (LINAC) based stereotactic radiotherapy (SRT) of uveal melanoma has been investigated for its potential as an alternative option (25–33). A reliable eye fixation technique with online control to ensure successful irradiation of the tumor volume has been

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